The Generalized Likelihood Ratio Test

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Key Concepts

1) The likelihood ratio is the optimal test of simple, binary hypotheses.

2) In general there is no optimal test for composite hypotheses, that is, when one or both of the hypotheses has unknown parameters. Examples of common unknown parameters are signal amplitude or noise power.

3) The generalized likelihood test (GLRT) is a principled strategy for generating a good test statistic.

4) The GLRT replaces the unknown parameters in the probability density functions (for each hypothesis) by their maximum likelihood estimates.

Video Screenshots

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Binary hypothesis test $T(\underline{x}) \underset{H_0}{\overset{H_1}{\gtrless}} \eta$ 1

Simple H_1, H_0: optimal $T(\underline{x})$ is likelihood ratio
gives maximum possible P_D for any given P_{FA}

Composite H_1, H_0 (unknown parameters in $f_{H_1}(\underline{x}), f_{H_0}(\underline{x})$)

In general the optimal test is unknown

Use "sensible" strategies

For example, generalized likelihood ratio test
(GLRT)

GLRT: replace unknown parameters in $f_{H_1}(\underline{x}), f_{H_0}(\underline{x})$ with maximum likelihood estimates

H_0: unknown parameters $\underline{\alpha}_0$ $f_{H_0}(\underline{x}; \underline{\alpha}_0)$

H_1: unknown parameters $\underline{\alpha}_1$ $f_{H_1}(\underline{x}; \underline{\alpha}_1)$

GLRT $$L(\underline{x}) = \frac{f_{H_1}(\underline{x}; \hat{\underline{\alpha}}_1)}{f_{H_0}(\underline{x}; \hat{\underline{\alpha}}_0)} \underset{H_0}{\overset{H_1}{\gtrless}} \eta$$

where $\hat{\underline{\alpha}}_i = \arg\left[\max_{\alpha_i} \left\{ f_{H_i}(\underline{x}; \alpha_i) \right\}\right]$ is the maximum likelihood estimate of α_i

Example: Known signal shape in noise

$\underline{x}: N \times 1$

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$H_0: \underline{x} = \underline{w}$ "noise" $\underline{x} \sim N(0, \sigma^2 \underline{\underline{I}})$, σ^2 unknown

$H_1: \underline{x} = A\underline{s} + \underline{w}$ "signal + noise" $\underline{x} \sim N(A\underline{s}, \sigma^2 \underline{\underline{I}})$ A, σ^2 unknown, $\underline{s}$ known

PDF's:

$$f_{H_0}(\underline{x}; \sigma^2) = \frac{1}{(2\pi\sigma^2)^{N/2}} \exp\left\{-\frac{1}{2\sigma^2} \underline{x}^T \underline{x}\right\}$$

$$f_{H_1}(\underline{x}; A, \sigma^2) = \frac{1}{(2\pi\sigma^2)^{N/2}} \exp\left\{-\frac{1}{2\sigma^2} (\underline{x} - A\underline{s})^T (\underline{x} - A\underline{s})\right\}$$

Find MLE's

$H_0: \hat{\sigma}_0^2 = \frac{1}{N} \underline{x}^T \underline{x}$

$H_1:$

$$\hat{A} = \underline{s}^T \underline{x} / (\underline{s}^T \underline{s})$$

$$\hat{\sigma}_1^2 = \frac{1}{N} (\underline{x} - \hat{A}\underline{s})^T (\underline{x} - \hat{A}\underline{s})$$

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GLRT

$$L(\underline{x}) = \frac{f_{H_1}(\underline{x}; \hat{A}, \hat{\sigma}_1^2)}{f_{H_0}(\underline{x}; \hat{\sigma}_0^2)} = \left(\frac{\hat{\sigma}_0^2}{\hat{\sigma}_1^2}\right)^{N/2} \exp\left\{-\frac{1}{2\hat{\sigma}_1^2}(\underline{x}-\hat{A}\underline{s})^T(\underline{x}-\hat{A}\underline{s}) + \frac{1}{2\hat{\sigma}_0^2}\underline{x}^T\underline{x}\right\}$$

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Since $\hat{\sigma}_1^2 = \frac{1}{N}(\underline{x}-\hat{A}\underline{s})^T(\underline{x}-\hat{A}\underline{s})$ and $\hat{\sigma}_0^2 = \frac{1}{N}\underline{x}^T\underline{x}$: $\exp\left\{-\frac{N}{2}+\frac{N}{2}\right\} = 1$

$$L(\underline{x}) = \left(\frac{\hat{\sigma}_0^2}{\hat{\sigma}_1^2}\right)^{N/2} \underset{H_0}{\overset{H_1}{\gtrless}} \eta \quad \Rightarrow \quad L'(\underline{x}) = \frac{\hat{\sigma}_0^2}{\hat{\sigma}_1^2} \underset{H_0}{\overset{H_1}{\gtrless}} \eta'$$

$$L'(\underline{x}) = \frac{\underline{x}^T\underline{x}}{(\underline{x}-\hat{A}\underline{s})^T(\underline{x}-\hat{A}\underline{s})} \xrightarrow{\text{algebra}} L''(\underline{x}) = \frac{(\underline{x}^T\underline{s})^2}{\underline{x}^T\underline{x} - (\underline{x}^T\underline{s})^2} \quad (\underline{s}^T\underline{s} = 1)$$

$$\underline{x}^T\underline{s} = \sum_{n=0}^{N-1} x[n]s[n] \quad \text{"matched filter"}$$

$$L''(\underline{x}) = \frac{\text{energy in } \underline{x} \text{ aligned with } \underline{s}}{\text{energy in } \underline{x} \text{ not aligned with } \underline{s}}$$

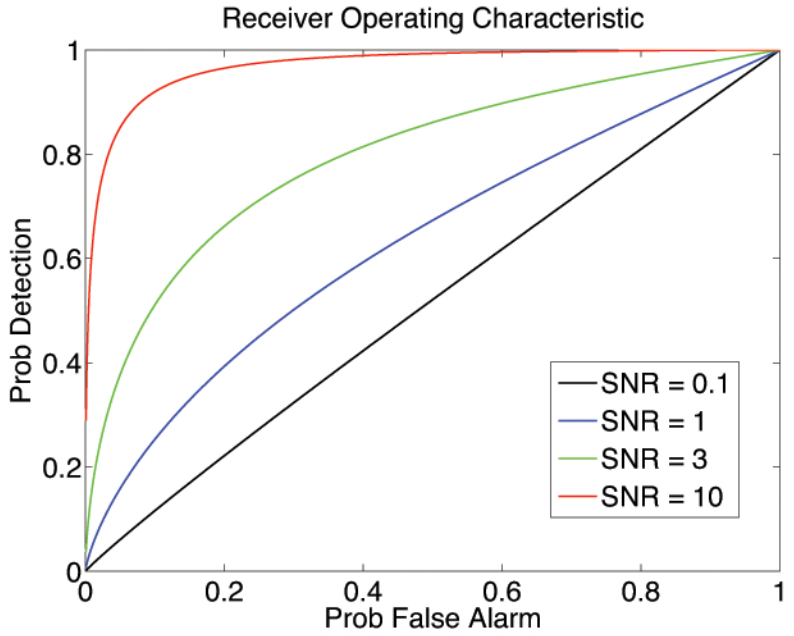
$L''(\underline{x})$ is equivalent to a ratio of indep. χ^2 random variables

H_0: $L''(x) \sim F_{1,N-1}$ $\quad H_1$: $L''(\underline{x}) \sim F_{1,N-1}(A^2/\sigma^2)$

Prob false alarm depends only on N, not σ^2 or A (CFAR)

Prob detection depends on N, SNR = A^2/σ^2

$N = 20$



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